

DATA*6700 – Project Report

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1. Abstract:

Environmental justice addresses the disproportionate exposure of marginalized communities to environmental hazards and toxic pollutants. This project leverages the **National Pollutant Release Inventory (NPRI)** database and the **Canadian Index of Multiple Deprivation (CIMD)** data to identify distinct kinds of overburdened communities based on their demographic characteristics and the types of pollutants released by nearby industrial facilities, rather than *ranking* such communities, as is done in the United States. The project focuses on NPRI data from 2007 to 2022, with a specific emphasis on Criteria Air Contaminants (CACs), which constitute 97% of pollutants released into the air by weight. The methodology involves three major stages: data pre-processing, clustering analysis, and results interpretation. Data pre-processing includes retrieving and normalizing NPRI data, which is then trimmed to relevant variables. The analysis phase employs k-means clustering to identify patterns across facility records. The optimal number of clusters - four - is determined using silhouette analysis. Results are interpreted by examining summary statistics and visualizing findings to understand each cluster's meaning. The findings provide valuable insights into the various kinds of communities overburdened by environmental hazards and the factors that differentiate them (i.e., different industries, pollutants, and demographic profiles). These insights are crucial for targeting social and economic policies such as Bill C-226 (“National Strategy Respecting Environmental Racism and Environmental Justice Act”). In conclusion, this project provides an analysis of environmental justice issues in Canada, offering actionable recommendations for policymakers and community organizations to address environmental racism and promote environmental justice. Future work will involve integrating additional data sources and employing advanced clustering techniques for more detailed analyses.

2. Introduction:

Environmental justice is concerned with addressing the disproportionate exposure of marginalized communities to environmental hazards and toxic pollutants. This inequality is often rooted in historical and societal factors that place these communities closer to sources of pollution such as highways, garbage disposal sites, and industrial facilities. Recent legislative efforts, such as Canada's "National Strategy on Environmental Racism and Environmental Justice Act" (Bill C-226), emphasize the need to study and address environmental racism, which involves examining the intersection of race, socio-economic status, and environmental risk. Although environmental racism is not a new concept, it has gained increased attention due to its serious impact on public health and social equality.

The studies by [6] Giang et al. (2020) and Jorquera and Villalobos (2020) provide valuable insights into the analysis of environmental hazards through innovative methodologies. Giang et al. (2020) utilized cumulative hazard indices (CHIs) to assess air pollution burdens across Canadian cities, revealing significant differences in exposure among marginalized communities - particularly Indigenous, low-income, and racialized groups - depending on the city. This study underscores the need for tailored policy interventions. Meanwhile, Jorquera and Villalobos (2020) combined cluster analysis of air pollution and meteorological data to estimate long-term, hourly source apportionment of PM_{2.5} (particulate matter < 2.5 micrometers) and PM₁₀ (particulate matter < 10 micrometers). Their approach offers a more detailed understanding of pollution sources, particularly in urban areas with diverse emission profiles [6] (Jorquera & Villalobos, 2020).

A comprehensive analysis of the usage of NPRI data in peer-reviewed environmental research over 25 years, from its inception in 1994 to 2019 has been highlighted in the study, 'Use of the

National Pollutant Release Inventory in Environmental Research: A Scoping Review’ [4] (A Berthiaume, 2021). The review identifies that NPRI data, primarily used for air pollutant studies, have been increasingly adopted in research, particularly for geospatial analyses, environmental monitoring, and predictive modeling. The study reflects on the potential for broader NPRI use in underexplored fields, such as indigenous issues and biomonitoring, while also pointing out the need for improved data quality and accessibility. However, without considering socioeconomic and demographic data, NPRI data alone is insufficient to completely understand the impact of pollution on communities. The research by [9] Kodros et al. (2024) provides key insights into cumulative exposures to environmental and socioeconomic risks in Milwaukee County, Wisconsin, USA and has been referenced to understand similar disparities in Canadian contexts. The Canadian Index of Multiple Deprivation (CIMD) [5] is extremely valuable in this context. The CIMD (2021) is a geographically based index developed to examine multiple aspects of social disadvantage across Canada. The CIMD offers detailed demographic information, helping to identify communities that are particularly vulnerable to environmental hazards.

2.1 Gap in Knowledge:

Despite the availability of NPRI and CIMD data, there are considerable gaps in the current research and the tools used to analyze these datasets. A key issue is the dependence on self-reported data, which can result in biases and inaccuracies, as industries might underreport their emissions or omit certain pollutants. Moreover, NPRI data often lacks the necessary spatial and temporal detail, making it challenging to capture short-term variations and localized pollution patterns. Existing data science tools frequently fall short when it comes to integrating environmental and demographic data to provide a comprehensive understanding of environmental injustice. Most research tends to focus on either pollution data or socio-economic data, with few studies combining

both to identify the most impacted communities. Additionally, traditional statistical methods may not be sufficient to detect complex patterns and relationships within large, multidimensional datasets, highlighting the need for advanced analytical techniques like machine learning to address these challenges [8] (Hino, Benami, & Brooks, 2018).

A further significant gap is the failure of many studies to properly account for the proximity between environmental hazards and nearby residential populations, with traditional methods like unit-hazard coincidence overlooking the precise locations of hazards relative to community boundaries but it also doesn't account for the reality of how hazards are spatially distributed across space [7] Mohai and Saha (2006). The US approach, using tools like EJ-Screen and CEJST, ranks communities on a scale from 1 to 50,000 based on social and environmental disadvantages. While this method provides detailed rankings, it may not be as practical for policymaking. Clustering communities into a smaller number of groups, such as four clusters, offers a clearer view of common issues and the factors driving those problems, making it easier to target interventions effectively. Machine learning techniques, such as k-means clustering, have shown potential in environmental monitoring but are underutilized in this specific context. K-means clustering can help identify patterns and group similar data points, making it a valuable method for analyzing large datasets with multiple variables. However, its application in environmental justice research is still limited, and there is a need for more studies that leverage these techniques to understand the distribution of environmental hazards and their impact on marginalized communities.

2.2 Project Objective:

This project aims to bridge these gaps by integrating advanced data analysis techniques with comprehensive environmental and demographic datasets, specifically through k-means clustering. By leveraging the comprehensive datasets provided by the National Pollutant Release Inventory

(NPRI) and the Canadian Index of Multiple Deprivation (CIMD), this study aims to identify and analyze clusters of communities based on pollutant releases and their socio-demographic characteristics. The focus is on Criteria Air Contaminants (CACs) data from 2006 to 2021, which accounts for most air pollutants by weight. Through k-means clustering, the project seeks to uncover complex patterns and relationships within the data, offering a more detailed understanding of the spatial distribution of environmental hazards. The findings are expected to reveal key insights into areas facing similar exposure risks and pressures, thereby informing targeted interventions and policies aimed at mitigating environmental injustice and promoting community resilience across Canada.

3. Methods:

3.1 Data Collection and Description:

National Pollutant Release Inventory (NPRI):

The National Pollutant Release Inventory (NPRI) is a crucial tool for tracking pollution across Canada. It collects data from all industries and accounts for pollutants released into the water, air, land and the management and transfer of these substances. Established in 1993, the NPRI helps monitor a wide range of pollutants, offering valuable insights into pollution trends and sources. Every year, by June, industries that meet specific reporting criteria must report their emissions to the NPRI. These criteria include thresholds set for each pollutant—only if an industry's emissions exceed these thresholds over the past year are they required to report to the NPRI. This system ensures that significant sources of pollution are tracked, while smaller, less impactful emissions may not be included. From 2001 to 2006, the number of facilities reporting to the NPRI increased by about 6,000, enhancing the data's quality and comprehensiveness. The NPRI data reveals important trends, including increases in pollutant releases across various environmental media. For

instance, during 2006 to 2021 emissions to water rose by 15,233 tonnes, and releases to land increased by 6,166 tonnes, due to activities in sectors such as wastewater treatment and air transportation [Government of Canada, 2023].

The Canadian Index of Multiple Deprivation (CIMD):

The Canadian Index of Multiple Deprivation (CIMD) is a geographically based index developed to analyze various dimensions of societal disadvantage across Canada. It is inspired by the 2006 Canadian Marginalization Index (CAN-Marg). The CIMD was created in collaboration with Statistics Canada, incorporating feedback from multiple divisions within the organization to ensure a comprehensive and accurate measure of deprivation. The CIMD is derived from the Canadian Census, with the most recent version being based on the 2021 Census data. It provides a snapshot of deprivation by summarizing data across small geographic areas (dissemination areas) into four key dimensions: Residential Instability, Economic Dependency, Ethno-Cultural Composition, and Situational Vulnerability. These dimensions allow researchers and policymakers to assess and address social inequalities effectively, particularly in areas related to health, education, and justice [Statistics Canada, 2021].

In this project we focus on the four CIMD indexes:

- **Residential Instability:** This index measures the tendency of people in a neighborhood to move frequently and includes factors like housing stability, such as the proportion of rented versus owned homes and the number of people living alone.
- **Economic Dependency:** It evaluates the reliance on non-employment income sources, focusing on the proportion of elderly residents (aged 65 and older) and the dependency ratio between the working-age population (15-64) and dependents (0-14 and over 65).

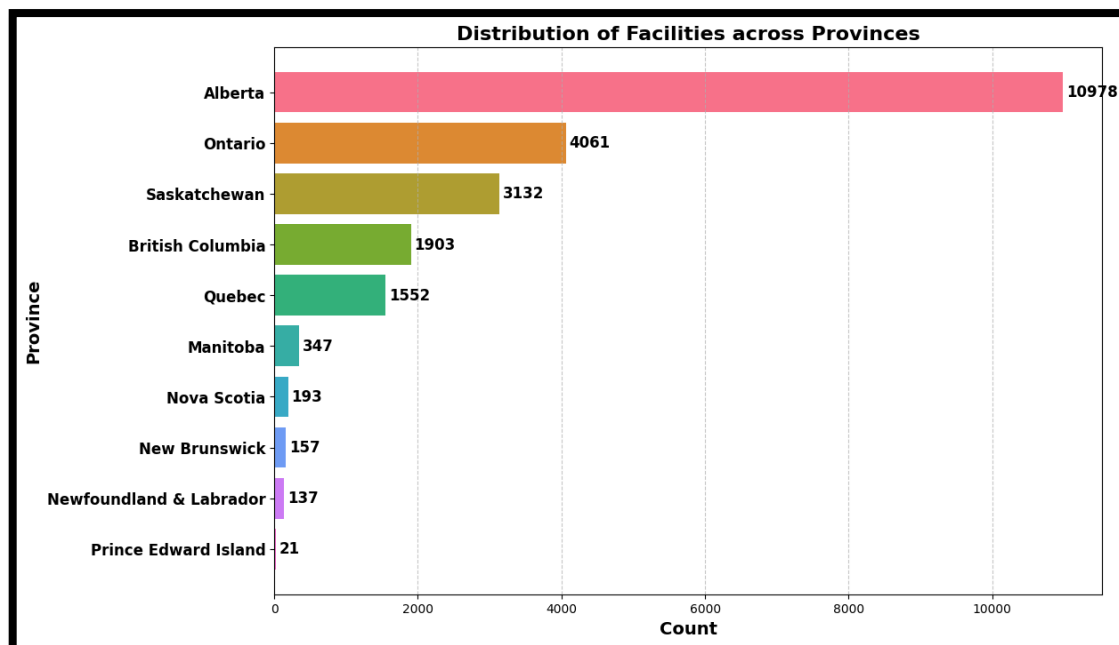
- **Ethno-Cultural Composition:** This index measures the community's cultural makeup by examining the proportion of recent immigrants, visible minorities, and individuals who do not speak either of Canada's official languages.
- **Situational Vulnerability:** It looks at socio-demographic factors such as education, housing, and other vulnerabilities, including the percentage of the population without a high school diploma, and those identifying as Indigenous.

3.2 Data Pre-Processing:

The data was loaded using the NPRI Python package, which includes data from 1993 to 2022. This package integrates the NPRI data with CIMD scores, providing pollutant values and corresponding CIMD scores along with geographic coordinates for all reporting industrial facilities. The integration process involved assigning CIMD data to each facility based on its Dissemination Area (DA), a geographic unit in Canada with a population between 400-700. A 5km buffer was created around each facility and the median CIMD score for each index was calculated from all DAs completely within that buffer; these scores were then assigned to the facility. After loading the data, the distribution of CIMD scores was observed to be normal, though some values were missing. For facilities lacking median CIMD scores, raw CIMD scores were assigned based on the DA they are located in. Pollutant values recorded in tonnes were converted to kilograms for better interpretability, and rows with missing NPRI identifiers were removed. Finally, the dataset was filtered to focus specifically on Criteria Air Contaminants (CACs) relevant to this analysis. The CACs consisted of the following pollutants: Carbon monoxide, Nitrogen oxides (expressed as nitrogen dioxide), Sulphur dioxide, Ammonia, Particulate matter (PM10 and PM2.5), and Volatile Organic Compounds (VOCs).

To determine the optimal set of features for this analysis, we first reduced the data to the most relevant features, while also experimenting with different timelines and sets of features. For example, we tested whether pollutant data spanning 15 years was more informative than using only the current year's data. Principal Component Analysis (PCA) on the resulting clusters revealed that data from the past 15 years (2006-2021) provided the most meaningful insights. This period not only aligns with the improved data quality mentioned earlier but also captures a wide historical range, allowing for the analysis of trends and variations over time. Based on these experiments, the dataset was refined by reducing the original 89 variables to 16 key features most relevant for assessing air pollution and its impact on the community. Additionally, the data was scaled down for clustering using the Min-Max method to account for outliers in pollutant values, ensuring that all features were on a comparable scale for accurate clustering analysis.

3.3 Data Visualization:



Figure

1: Distribution of Facilities across Provinces

In Figure 1, the bar chart depicts the distribution of industrial facilities across Canadian provinces, with Alberta emerging as the province with the highest number of facilities, with 10,978 facilities. This dominance is followed by Ontario and Saskatchewan, which host 4,061 and 3,132 facilities, respectively. The chart underscores the concentration of industrial activity in these regions, reflecting their significant roles in Canada's industrial landscape. In contrast, provinces like Prince Edward Island, with only 21 facilities, demonstrate a minimal industrial footprint, highlighting the substantial regional disparities in industrial presence across the country. These differences may have implications for regional economic activity, environmental impacts, and resource allocation, making the understanding of such distribution critical for policymakers and stakeholders.

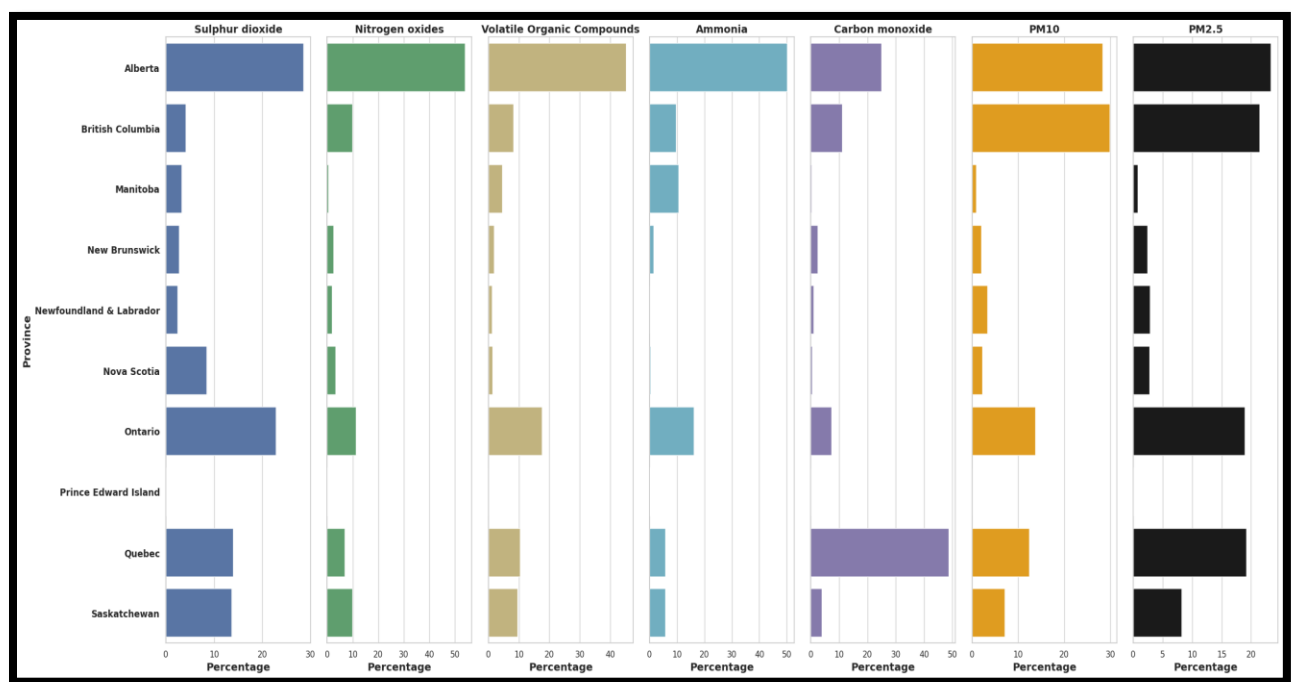


Figure 2: Distribution of all the 7 CACs across 10 Provinces in Canada

In Figure 2, the spread of the seven CACs across Canadian provinces is depicted, with each pollutant's contribution represented as a percentage of the total emissions across Canada. Alberta emerges as the leading province in the release of most pollutants, significantly contributing, for example, 54% of the total nitrogen oxides emissions in the country. British Columbia, on the other hand, shows dominance in particulate matter. Quebec stands out as the largest emitter of carbon monoxide. This visualization underscores the regional differences in pollutant contributions, reflecting the varied industrial activities and environmental impacts across the provinces.

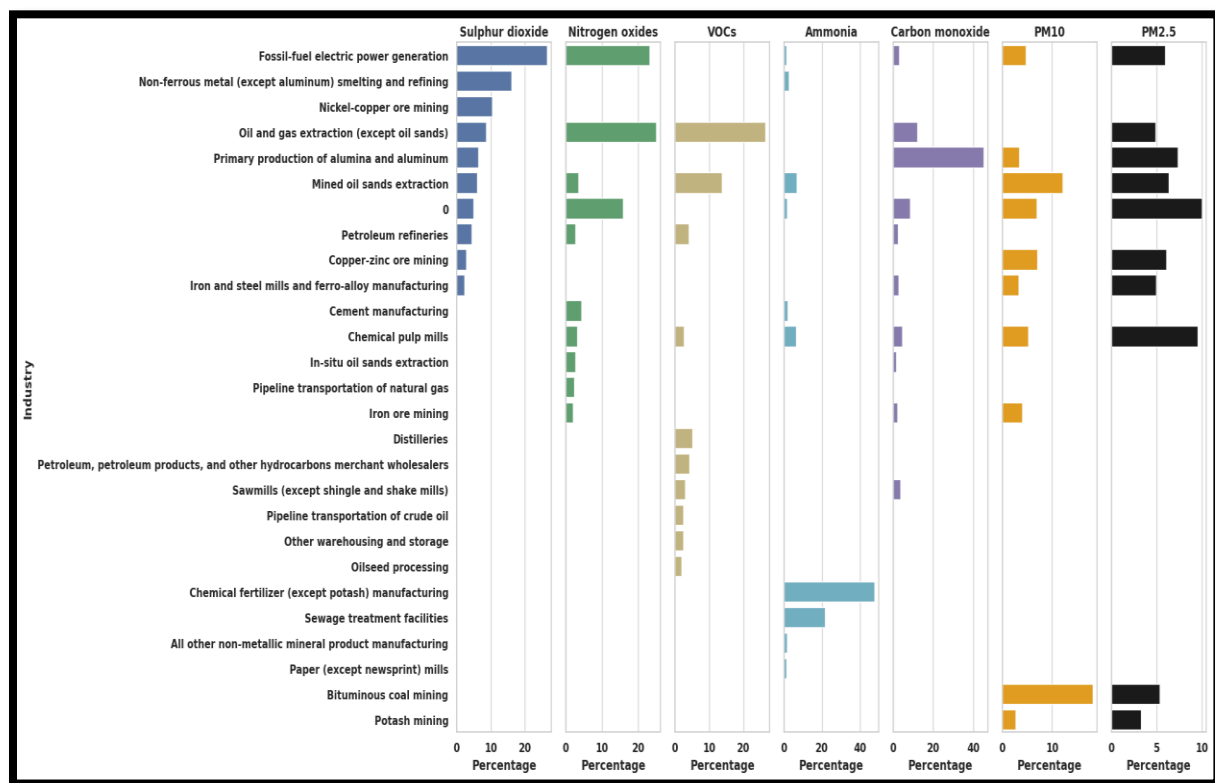


Figure 3: Distribution of CACs Across Industry types

In Figure 3, the bar charts illustrate the percentage contributions of various industries to the release of different CACs. Fossil-fuel electric power generation and oil and gas extraction are leading

sources of Sulphur dioxide and Nitrogen oxides, respectively. VOCs are emitted by the petroleum refining industry, while ammonia emissions are attributed to sewage treatment facilities. This visualization highlights the dominant polluting industries for each CAC, highlighting the significant role of specific sectors in national pollutant levels.

3.4 Clustering Analysis:

In this project, we utilized the k-means clustering algorithm to identify and group communities based on facility pollutant releases and demographic characteristics. The primary objective is to create clusters that could help in understanding how different communities are impacted by environmental hazards, thus providing insights into environmental justice.

k-means clustering:

K-means clustering is a widely used partitioning method in data analysis that aims to divide a dataset into a specified number of clusters (k), where each data point belongs to the cluster with the nearest mean value. The algorithm operates iteratively to assign data points to one of the k clusters by minimizing the variance within each cluster, which helps in identifying natural groupings within the data. K-means clustering is particularly beneficial in analyzing spatial data because it allows for the identification of regions or areas that share similar characteristics. k-means helps in visualizing patterns that might not be directly apparent. This is crucial for spatial analysis in environmental studies, where understanding the geographic distribution of pollution and its impact on communities is essential. Its simplicity allows for easy interpretation of the results. Unlike hierarchical clustering, which can become computationally intensive as the dataset grows, k-means is scalable and can handle substantial amounts of data with low computational cost. Additionally, k-means is effective in identifying spherical clusters, which are common in spatial data, and can easily be integrated with other spatial analysis tools.

In this project, we employed both the Elbow Method and Silhouette Analysis to determine the optimal number of clusters (k). While the Elbow Method helps identify the point where adding more clusters would reduce the quality, it does not directly evaluate the quality of cluster formation. Silhouette Analysis, however, provides a more detailed assessment by measuring both the cohesion within clusters and the separation between them. It offers a visual representation of how well each data point fits within its cluster and how distinct the clusters are from each other. Silhouette coefficient values range from -1 to $+1$. Negative values indicate poorly formed clusters, where data points may be misassigned. Higher positive values suggest well-formed clusters, with data points closely matched to their own clusters and distinct from others.

Finally, after evaluating the results, we proceeded with clustering. The cluster labels were then integrated back into the raw data, allowing for a more detailed analysis and interpretation of the information within each cluster. This step was crucial for inferring meaningful insights from the clustered data and understanding the distribution and characteristics of the facilities and pollutants across different clusters. We also observed that nearly one-third of the data consisted of facilities that reported zero emissions for all seven CACs. To enable comparison and gain insights into how these facilities compare against those with reported emissions, we grouped these facilities into a separate cluster and added them back to the data. In the following section, we will present and discuss the results of our analysis, highlighting the key findings and insights derived from the clustered data.

4. Results:

4.1 Prose Summary:

The results from the Silhouette analysis revealed that the optimal number of clusters (k) for this study is 4. With this value, clustering was performed, and the cluster labels were integrated back into the raw data. A bar chart was then plotted to illustrate the distribution of industries within each cluster, revealing the predominant types of industries in each group. Following this, spatial plots of the clusters were created using the geographic coordinates of the facilities, highlighting the regional concentration of each cluster across Canada. To further analyze the clusters, a summary statistics table was generated, aggregating the features based on cluster labels. This table allowed us to identify the most dominant pollutants and corresponding CIMD scores within each cluster. By synthesizing these insights, we defined and described the characteristics of each cluster, providing a comprehensive understanding of the environmental and demographic factors that distinguish them.

4.2 Silhouette Analysis Results:

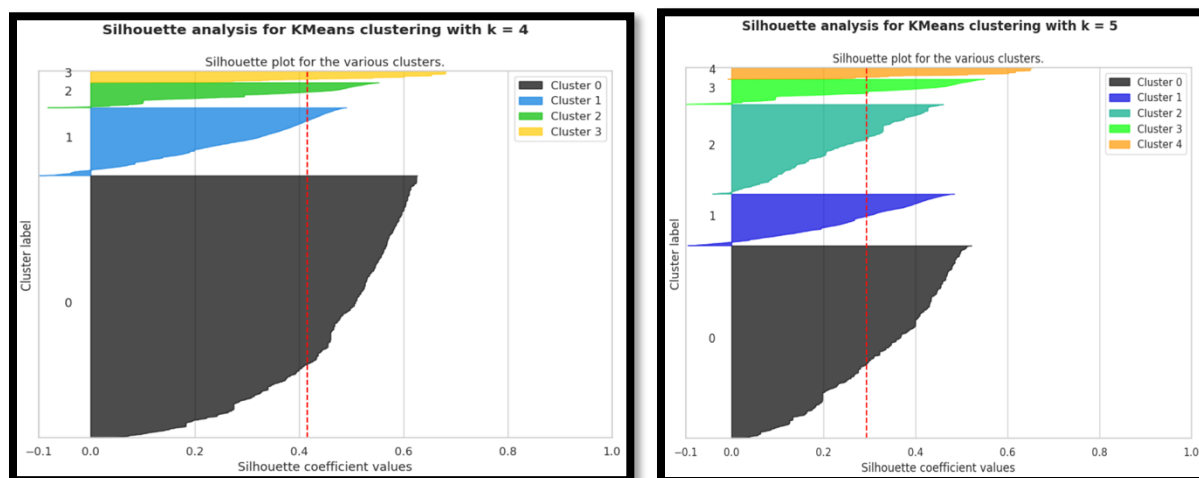


Figure 4 & 5: Silhouette Analysis for $k=4$ & $k=5$, respectively.

Figures 4 and 5 show Silhouette Analysis plots for k-means clustering with $k=4$ and $k=5$, respectively. Each plot represents the silhouette coefficient for every point in the dataset, grouped by their assigned cluster. In the plot for $k=4$, we observe that the clusters have high silhouette coefficients of 0.42, particularly for Cluster 0, which shows a strong separation from the other clusters. The plot for $k=5$, however, shows a more scattered distribution of silhouette coefficients of 0.29, indicating less distinct clusters. The red vertical indicates the Silhouette coefficient value, wherein for the plot (figure 4) $k=4$, the co-efficient value is higher comparatively, and as the k value further increases the coefficient value decreases. Notably, while the silhouette plot for $k=4$ does show some negative values, they are more prominent in the $k=5$ plot, particularly in Clusters 1 and 2. This suggests that certain facilities in $k=5$ may not have been correctly assigned, due to overlap or ambiguity in cluster boundaries. These negative values indicate that some points might be closer to neighboring clusters, raising concerns about clustering accuracy. Therefore, $k=4$ was chosen as the optimal number of clusters, offering a more distinct and well-defined clustering solution.

4.3 Top 10 Industries in Each Cluster:

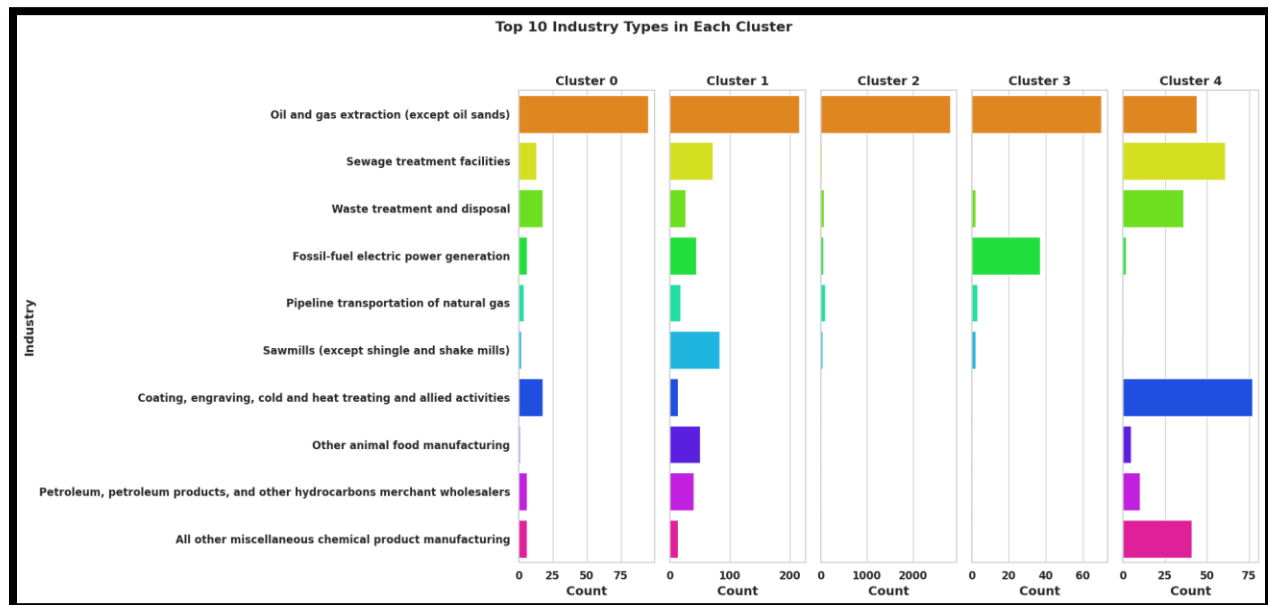


Figure 6: Top 10 Industries in Each Cluster

This bar chart illustrates the distribution of the top 10 industry types across the five clusters identified in the analysis. It is important to note that the x-axes differ for each cluster, this variation highlights how some clusters are more heavily dominated by certain industries than others. Each pollutant's percentage contribution is calculated across the entire country.

- **Cluster 0:** In this cluster, the predominant industries include Oil and Gas, Waste Treatment, and Coating sectors.
- **Cluster 1:** Distinct from other clusters, it consists of Sawmills (except shingle and shake mills), and Sewage treatment facilities.
- **Cluster 2:** This large cluster of facilities mostly consists of those in the Oil and Gas Extraction industry, like Cluster 0.

- **Cluster 3:** This cluster has a strong presence of Fossil-Fuel Electric Power Generation, Oil and Gas Extraction industry.
- **Cluster 4:** It is dominated by Coating, Engraving and Heat-treating Industries and Sewage Treatment facilities.

Cluster 4 represents the additional cluster that was added to include facilities that have reported null emissions for all Criteria Air Contaminants (CACs), providing a basis for comparison with other clusters that have reported emissions. Therefore, this bar chart provides valuable insights into the industrial landscape across different clusters, highlighting the dominant sectors within each group. By analyzing the distribution of industries, we can infer the potential environmental impacts associated with each cluster and identify which industries contribute most significantly to pollutant emissions.

4.4 Small Multiples:

In this section, we will explore the spatial distribution of clusters by visualizing the geographic concentration of facilities within each cluster, referred to as small multiples. This analysis will provide a visual representation of how different clusters are spread across Canada, offering insights into regional patterns of industrial activity and environmental impact.

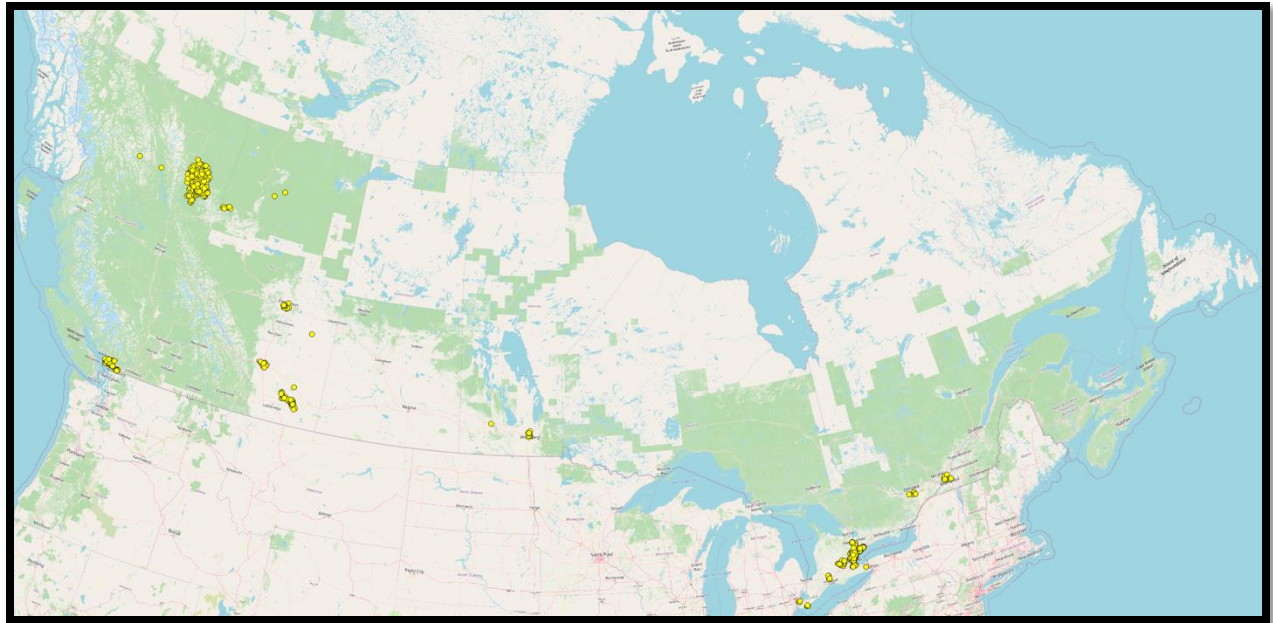


Figure 7: Cluster 0

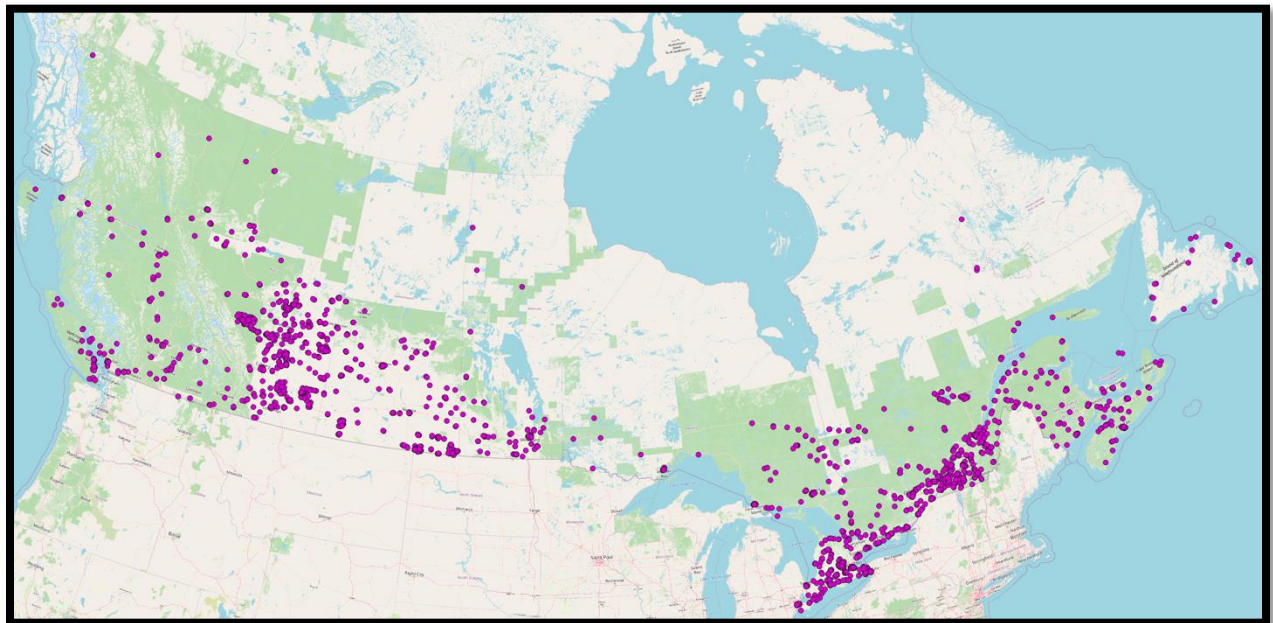


Figure 8: Cluster 1

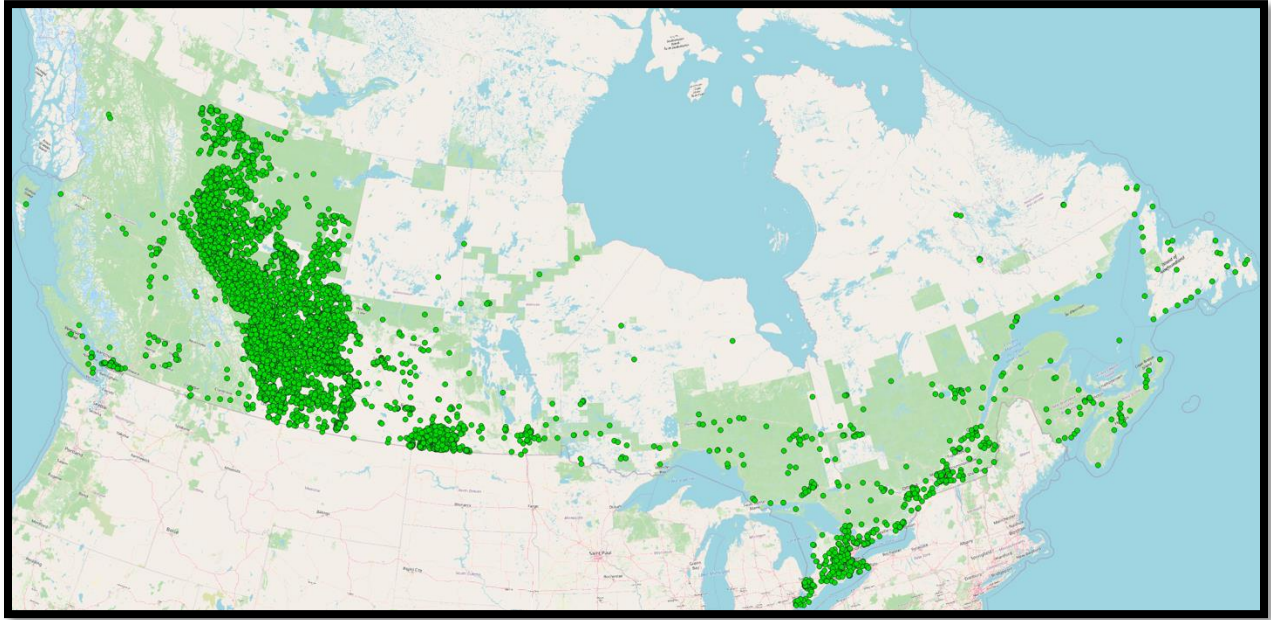


Figure 9: Cluster 2

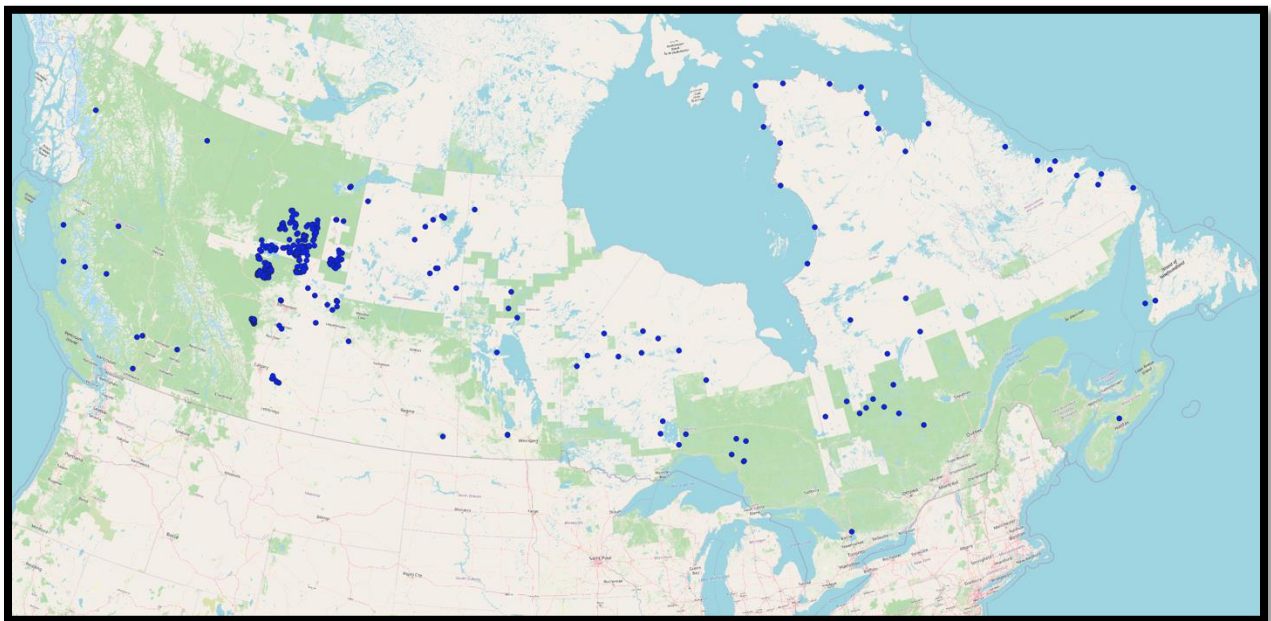


Figure 10: Cluster 3

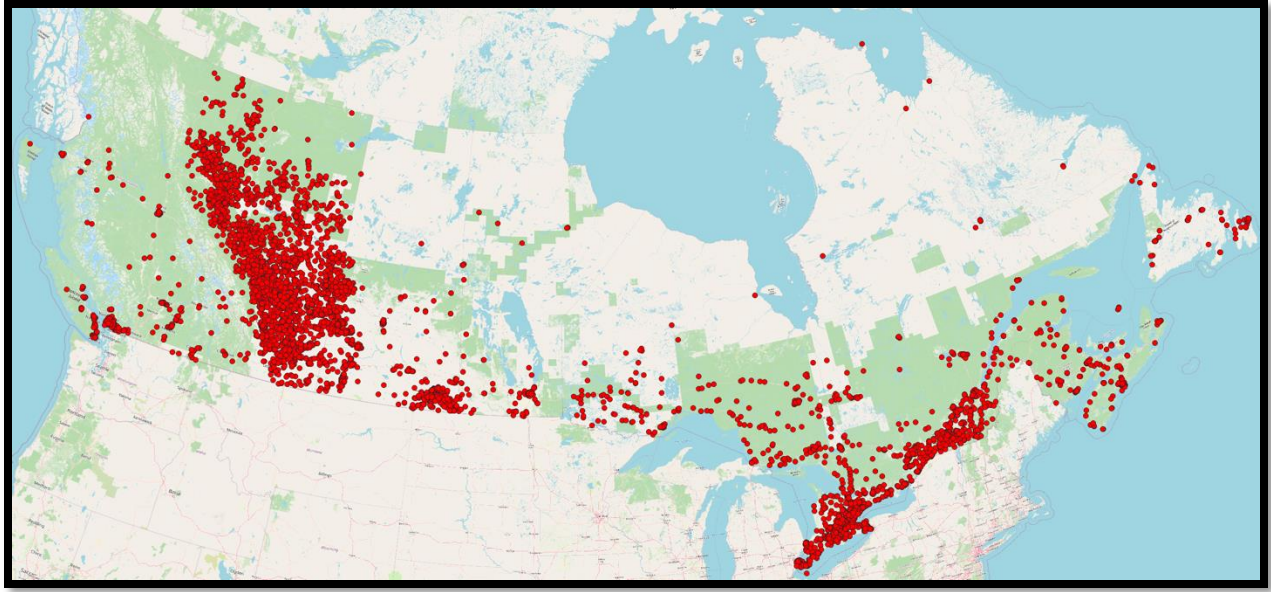


Figure 11: Cluster 4

In the above-depicted figures, we analyze the geographic distribution of clusters across Canada.

- **Cluster 0:** This cluster (figure 7) is in the northern regions of British Columbia and in major Canadian cities such as Toronto, Vancouver, Calgary, and Montreal, as well as other significant urban centers across Canada.
- **Cluster 1:** This cluster (figure 8) is more evenly distributed across various provinces, indicating a mix of industries without a strong geographic concentration.
- **Cluster 2:** This is the largest cluster of all (figure 9), like Cluster 0, this also has a noticeable concentration in the western provinces, particularly Alberta, but with a more scattered distribution overall.
- **Cluster 3:** This cluster (figure 10) has its presence mostly in the northern regions of Canada.
- **Cluster 4:** This is the additional cluster (figure 11), this cluster comprises of various kinds of industries which have reported null emissions for all the 7 CACs it is evenly spread across Canada.

4.5 Summary Statistics:

Cluster	Carbon Monoxide	Ammonia	Sulphur Dioxide	PM10	PM2.5	Nitrogen Oxides	VOCs	Median Instability_2021	Median Dependency_2021	Median Composition_2021	Median Vulnerability_2021	
0	238,557	21,593	132,839	35,318	19,122	295,746	17,268	-0.56	1.17	-0.27	0.29	Notable CIMD Scores
1	1,965,168	47,066	2,339,285	187,943	103,854	776,663	24,436	0.29	-0.46	0.04	0.17	Notable Pollutant Values in Kg
2	629,493	10,942	661,608	113,964	34,399	539,612	10,097	-0.84	-0.76	-0.40	0.07	Highest Values in Each Column
3	348,006	1,672	85,575	240,782	66,903	436,620	26,024	-0.47	-0.41	0.31	4.36	
4	0	0	0	0	0	0	0	-0.39	-0.27	-0.18	0.11	

Table 1: Summary of the Clusters

Table 1 presents summary statistics, where all the features have been aggregated based on the cluster labels. The green boxes in each row highlight the most dominant pollutant within that cluster. The red boxes represent the most dominant CIMD scores in each cluster, reflecting the significant social deprivation factors.

For example, in cluster 0, the dominant pollutant is Nitrogen Oxide, and the corresponding notable CIMD score is High Economic Dependency. The blue boxes across each column indicate the highest values among all clusters, highlighting which cluster has the peak value for each feature. For instance, Carbon Monoxide has the highest value in Cluster 1, while PM10 reaches its peak value in Cluster 3.

This table is significant as it provides a clear overview of the primary pollutants and social deprivation scores within each cluster, enabling a better understanding of the environmental and demographic characteristics that define each cluster.

4.6 Clusters Interpretation:

Clusters	Predominant Industry	Geographic Concentration	Description
0	Waste Treatment and Oil and Gas	• Northern BC and in Major Cities	• Nitrogen Oxides and Carbon Monoxide. • Many people in this cluster rely on government assistance and social support, reflecting economic hardship.
1	Sewage Treatment and Sawmills	• Clustered across Canada	• Sulphur Dioxide. • Consists of transient populations, which can impact community cohesion and access to services.
2	Oil and Gas	• Clustered across Canada	• Sulphur Dioxide and Carbon Monoxide. • This Cluster has a relatively low financial dependency, more stability, and less diversity compared to other clusters.
3	Fossil Fuel Electric Power Generation	• Clustered in Northern regions of Canada	• Nitrogen Oxides. • Comprises of indigenous people, low educational attainment, thus making communities more vulnerable to adverse conditions.
4	Coating and Chemical Product Manufacturing	• Clustered across Canada	• Null emissions. • The population in this cluster has a relatively low economic dependency, but there is also evidence regarding situational vulnerability.

Table 2: Clusters Description

Table 2 provides a structured interpretation of the five clusters identified in the analysis by organizing key information into three columns: Predominant Industry, Geographic Concentration, and Description. The "Predominant Industry" column identifies the main industry types within each cluster, while the "Geographic Concentration" column highlights the general locations where these clusters are primarily found across Canada. The "Description" column offers insights into the notable environmental and social characteristics of each cluster, such as the dominant pollutants and associated CIMD scores. For example, Cluster 0 is dominated by Waste Treatment and Oil and Gas industries, primarily located in Northern British Columbia and major cities. This cluster shows high emissions of Nitrogen Oxides and Carbon Monoxide, with populations often facing economic hardship. Each cluster similarly has distinct industries, geographic distributions,

and environmental and social characteristics, facilitating a comprehensive comparison across regions in Canada.

What distinguishes each cluster is not only the industry type or geographic location but also the unique environmental and social profile that emerges from the clustering. For instance, Cluster 1 stands out with its high Sulphur Dioxide emissions, making it distinct compared to other clusters, despite Sulphur Dioxide not being the most prominent pollutant overall. What sets Cluster 2 apart is its low financial dependency and higher stability compared to other clusters, making it a cluster where the environmental impact is significant, but the social resilience is stronger, which might influence how these communities cope with environmental challenges. Similarly, Cluster 3 is notable not just for its Nitrogen Oxides but because it has the highest PM10 and VOC values across all clusters, which makes it particularly important in the context of environmental risk assessment. Cluster 4 is unique due to the absence of emissions, combined with a low economic dependency. The results of the clustering analysis directly align with the project's objective by clustering industries and their associated emissions with demographic factors such as economic dependency and social cohesion, we have uncovered distinct patterns that reveal how environmental burdens and social deprivation are distributed geographically. These insights provide a clearer understanding of which communities are most affected by industrial activities.

5. Discussion:

5.1 Correlation of Findings with Project Objective:

The results of this analysis directly align with the initial objectives, which aimed to uncover the spatial distribution of environmental hazards across Canadian communities using advanced data analysis techniques. The application of k-means clustering effectively identified distinct clusters

based on facility pollutant releases and demographic characteristics. The use of Silhouette Analysis ensured that the optimal value of k was selected, allowing for a meaningful clustering of communities. These clusters revealed significant patterns, such as the concentration of certain pollutants in specific regions and the correlation between industrial activity and social deprivation. Challenges, such as handling missing data and integrating diverse dataset were successfully navigated, demonstrating the significance of the data science pipeline. Overall, the tools and methodologies employed provided valuable insights, meeting the project's objectives, and highlighting areas for potential intervention to address environmental justice concerns in Canada.

5.2 Comparison with Existing Studies:

The findings from this study significantly contribute to the existing literature on environmental justice and pollutant exposure, with a particular focus on the Canadian context. Prior research, such as the studies conducted by [6] Giang et al. (2022) and [4] Berthiaume (2021), has underscored the disproportionate exposure of marginalized communities to environmental hazards. These studies emphasize the critical need for more sophisticated spatial analyses to thoroughly understand these disparities. However, much of existing research has been limited by either focusing narrowly on specific pollutants or lacking a comprehensive integration of social vulnerability data. The findings of this study resonate with these earlier works by confirming the unequal burden of environmental hazards on vulnerable populations. However, they also advance the field by providing a more granular and detailed analysis of how industrial activities intersect with social vulnerabilities across different regions in Canada.

In addition to expanding upon Canadian-based research, this study's findings also align with and extend the work of [7] Mohai and Saha (2006), who explored the intersections of race, socioeconomic status, and environmental exposure in the United States. By applying similar

analytical frameworks to the Canadian context, this research not only reinforces the idea that marginalized communities are often disproportionately impacted by industrial pollution, but also introduces new insights specific to the pollutants and social factors present in Canada. The integration of the National Pollutant Release Inventory (NPRI) data with the Canadian Index of Multiple Deprivation (CIMD) scores through k-means clustering allows for a more comprehensive understanding of how these factors intersect. This approach, which focuses on the cumulative impact of multiple pollutants across various regions, addresses a significant gap in the literature by moving beyond analyses of individual pollutants or sectors. This detailed approach reveals complex interactions that were not fully explored in previous studies, highlighting the importance of combining environmental and socio-economic data to gain a complete understanding of environmental injustice.

5.3 Caveats and Limitations of the Study:

While this study offers important insights into the relationship between industrial pollution and social vulnerability in Canada, it does have certain limitations that warrant careful consideration. One primary limitation is the reliance on self-reported data from the NPRI. This data source, while comprehensive, may introduce biases or inaccuracies due to underreporting, inconsistent reporting practices by industries, or errors in data entry, which could affect the overall accuracy and reliability of the findings. Additionally, the integration of the Canadian Index of Multiple Deprivation (CIMD) scores, although valuable for identifying vulnerable communities, may not fully capture the multifaceted nature of social vulnerability. The CIMD is based on broad geographic areas, and thus, it might overlook more localized variations within communities that could be significant in understanding the full scope of environmental injustices.

While the k-means clustering methodology used in this study is effective in identifying distinct patterns, it has limitations that must be acknowledged. This method can oversimplify the complex relationships between variables, potentially overlooking subtle but important nuances. Clusters formed may not fully capture the intricate dynamics between environmental and social factors, particularly in regions with sparse or highly variable data. Additionally, the study's focus on Criteria Air Contaminants (CACs) narrows the scope, as it excludes other significant pollutants like water and soil contaminants or toxic air pollutants. This limitation may result in a partial view of the environmental challenges faced by communities. Moreover, the geographical focus on Canada, while relevant for this context, may limit the applicability of the findings to other regions with different industrial practices and regulatory frameworks. Future research should aim to overcome these challenges by incorporating a broader range of data, pollutants, and more sophisticated analytical techniques to achieve a more comprehensive understanding of environmental justice issues.

5.4 Next Steps in Research:

Future research could significantly expand this study by exploring several key areas.

- Clustering places or regions with similar analyses could provide more localized insights, helping to identify specific areas most affected by environmental hazards.
- Enhancing the data used in the analysis by integrating additional data sources would improve the study's overall quality and comprehensiveness. For example, incorporating health data could offer a distinct perspective, linking environmental factors with public health outcomes.

- Additionally, expanding the scope to include pollutants beyond Criteria Air Contaminants (CACs), such as water and soil contaminants, could provide a more holistic view of environmental risks.
- Temporal analysis would be another valuable addition, offering insights into how these risks and their impacts evolve over time.
- Finally, employing advanced machine learning techniques could refine the clustering analysis, allowing for the identification of more complex patterns and interactions between environmental and social factors, leading to more actionable insights for policymakers and communities.

5.5 Concluding Remarks and Broader Significance:

Here are the concluding remarks and broader significance presented:

- Conducted clustering analysis to identify key industries and geographic areas contributing to environmental injustice in Canada.
- By combining the National Pollutant Release Inventory (NPRI) with the Canadian Index of Multiple Deprivation (CIMD), derived important insights by examining the clusters and its corresponding attributes.
- This study highlights the importance of using detailed data analysis to create policies that effectively address environmental justice issues.
- The analysis underscores the importance of empowering communities through education and skill-development initiatives, enabling them to actively participate in addressing and mitigating the environmental risks they face.
- Implementing social support and economic development programs in vulnerable communities is recommended to enhance their resilience against environmental hazards.

- The project recommends encouraging standardized and transparent reporting practices among industries to improve data quality and reliability.
- The study demonstrates the value of collaboration between environmental agencies, social services, and community organizations to address the complexity of environmental justice issues.

6. References:

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